

Weekly Quiz 5

1. True or False?

1. In the fully observed case where we want to learn potentials $\phi_C(y_C)$ for each clique, with no prior on the distribution of the potentials, the maximum likelihood of the data given θ is achieved when $P(Y_i = y_i | Y_{Pa(i)} = y_{Pa(i)}) \propto \text{Number of occurrences of } y_i, y_{Pa(i)} / \text{Number of occurrences of } y_{Pa(i)}$.
2. In class, a variational approximation was used for

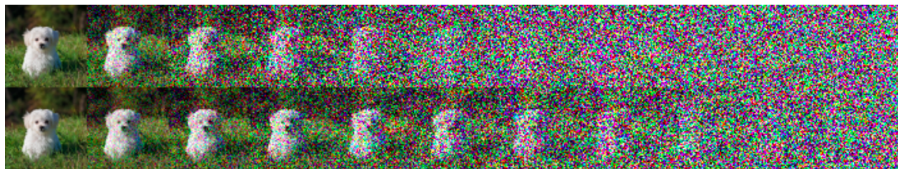
$$\max_{\theta} \sum_{i=1}^N \log \int_z P_{\theta}(\mathbf{x} | z) P_{\theta}(z) dz$$

by introducing an appropriate $q_{i,z}$. The function which is being maximized can be computed at any q, θ in its domain as

$$\sum_{i=1}^N \int_z \left(q_{i,z} \log P_{\theta}(\mathbf{x}^i | z) P_{\theta}(z) - q_{i,z} \log q_{i,z} \right) dz$$

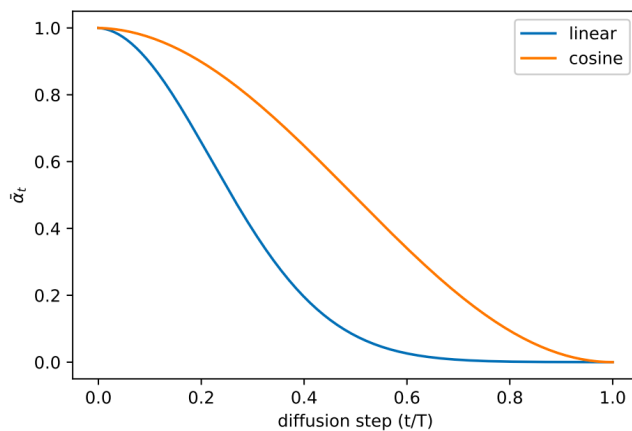
Note that you can assume there are no errors in the statement or formulas until now.

For any dataset, setting $q_{i,z} = P_{\theta}(\mathbf{x}^i | z) P_{\theta}(z)$ makes the above approximation 0, hence the approximation is always non-negative.



3.

The above image shows how noise is added according to two different schedules which are unknown. The variation of $\bar{\alpha}_t$ with time for two schedules, linear and cosine, are shown below. Based on the images and the graph, it is more likely that the top series of pictures represent noise added according to a cosine schedule, and the bottom set of pictures represent noise added according to a linear schedule. True or False?



4. If for some function $f(x)$ such that $\log f$ is concave over $[0, 1]$, it is the case that $\tilde{\alpha}_t = f\left(\frac{t}{T}\right)$ for $t = 0, 1, \dots, T$, then the β_t values are always non-decreasing for increasing t .

Solution:

1. True. The first observation is the global decomposability of the log likelihood function in a BN, where the log likelihood of the data given the parameters is a sum of log likelihood functions at every variable given its parents. For a clearer explanation, you can refer to Section 17.2.2 of the Koller-Friedman book. Now, we need to optimize each of the local loss functions individually. The local log-likelihood is

$$\ell_i(\theta_{X_i|\text{Pa}(X_i)}; D) = \sum_{x_i, \text{pa}_i} N(x_i, \text{pa}_i) \log \theta_{x_i|\text{pa}_i}.$$

Maximizing subject to $\sum_{x_i} \theta_{x_i|\text{pa}_i} = 1$ gives

$$\hat{\theta}_{x_i|\text{pa}_i} = \frac{N(x_i, \text{pa}_i)}{\sum_{x'_i} N(x'_i, \text{pa}_i)}.$$

I.e each $P(Y_i = y_i | Y_{\text{Pa}(i)} = y_{\text{Pa}(i)}) = \text{Number of occurrences of } y_i, y_{\text{Pa}(i)} / \text{Number of occurrences of } y_{\text{Pa}(i)} \text{ in the data.}$

2. False. There are constraints that $\sum_z q_{i,z} = 1$, or equivalently $\int_z q_{i,z} = 1$
 3. False. As we can see in the picture, the $\bar{\alpha}_t$ values reduce much faster under the linear schedule, meaning that the image devolves to pure noise much faster (in fact, the last 25% or so of the process adds virtually no useful information, instead converting near-random noise to noise. In comparison, the $\bar{\alpha}_t$ values are higher under the cosine schedule for longer).
 4. True. $\beta_t = 1 - \frac{\tilde{\alpha}_t}{\tilde{\alpha}_{t-1}}$, $\beta_{t-1} = 1 - \frac{\tilde{\alpha}_{t-1}}{\tilde{\alpha}_{t-2}}$. $\beta_t \geq \beta_{t-1} \leftrightarrow \log(\tilde{\alpha}_{t-2}) + \log(\tilde{\alpha}_t) \leq 2 \log(\tilde{\alpha}_{t-1})$, which is satisfied if $\log f$ is concave.
2. The KL divergence $D_{KL}(p, q)$ of two univariate Gaussians $p \sim \mathcal{N}(\mu_1, \sigma_1)$, $q \sim \mathcal{N}(\mu_2, \sigma_2)$ is given as $\log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$. The VAE loss can be expressed as $\text{loss}(\theta, \phi) = \sum_{i,r} -\frac{1}{r} \log \mathcal{N}(x_i^r | \mu_{x^i|z}^r, \sigma_{x^i|z}^r) + \sum_i \mu_{z|x^i}^2 + \sigma_{z|x^i}^2 - \log \sigma_{z|x^i}$.
1. If $\tilde{\alpha}_t$ does not tend to 0 as the number of timesteps taken (t) tends to T, then the term $D_{KL}(q(x_T|x_0), P(x_T))$ need not approach its theoretical minima of 0, even if $x_0 = 0$. True or false?
 2. To train the denoiser, it was determined that one could run gradient descent on an MSE loss which is a function of ϵ_θ , as

$$L_{t-1} = \mathbb{E}_{x_0, \epsilon} \left[\frac{\beta_t^2}{2\sigma_t^2(1 - \beta_t)(1 - \bar{\alpha}_t)} \|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right] + C$$

Note that you can assume there is no error in the above statement. A well-trained noise prediction network can be used to compute the mean $\mu_\theta(x_{t-1} | x_t)$ of $P_\theta(x_{t-1} | x_t)$. When sampling from $N(\mu_\theta(x_{t-1} | x_t), \sigma_t^2)$, it is better to set σ_t very high as it does not affect the loss function and can introduce better diversity to the responses. True or False?

3. To train the noise prediction network, one must sample x_0 and then sequentially generate $x_1, x_2 \dots x_T$ by adding noise.
4. Just like in a DDPM, where the L_T term $D_{KL}(q(x_T|x_0) || p(x_t))$ can be ignored since its optimal parameters are known beforehand, the encoder loss in a VAE can be optimized beforehand by setting $\mu_{z|x_i} = 0$, $\sigma_{z|x_i} = 1$

5. The VAE loss, given above, cannot be decomposed into a function of only the encoder parameters from the previous iteration + a function only of the decoder parameters from the previous iteration.

Solution:

1. True. Consider the univariate case, where we would have the KL divergence to be $-\log \sqrt{(1 - \tilde{\alpha}_t)} + \frac{\tilde{\alpha}_t(x_0)^2 - \tilde{\alpha}_t}{2}$. The required result can be seen by plotting $\frac{-\log(1-x)-x}{2}$.
2. False. The variance of P_θ factors into the C term in the loss above. The univariate case once again gives a simple example: setting σ_2 to be very high would blow up the $\log \sigma_2$ factor, while perhaps getting the $\frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2}$ term to reduce. But as you can see if you plot a graph of $\log x + \frac{1}{2x^2} - \frac{1}{2}$ (which assumes we have somehow exactly matched the means), the loss is least when $\sigma_1 = \sigma_2$. σ_1 is known beforehand, dependent on the sampling technique. When x_0 is deterministically set to a point, $\sigma_2^2 = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$ is the theoretical variance of $q(x_{t-1} | x_t, x_0)$. The proof is outside the scope of the course, but, as mentioned in the slides, can be achieved by solving for $q(x_{t-1} | x_t, x_0)$ using the equations

$$q(x_t | x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1 - \beta_{t-1}} x_{t-1}, \beta_{t-1} \mathbf{I}\right)$$

$$q(x_{t-1} | x_0) = \mathcal{N}(x_{t-1}; \sqrt{\bar{\alpha}_{t-1}} x_0, (1 - \bar{\alpha}_{t-1}) \mathbf{I})$$

Setting $\sigma_2^2 = \beta_t$ has been claimed to be the theoretical optima for when $x_0 \sim \mathcal{N}(0, 1)$ (section 3.2 of ‘Denoising Diffusion Probabilistic Models’)

3. False. Since the forward process is a fixed Markov chain, we can easily carry out only the sampling step, and then skip to an arbitrary timestep. This might even be beneficial if the number of timesteps is very large and one can then train the noise predictor on a few timesteps per image rather than having to commit to 1000s of timesteps per image and being forced to train on fewer images due to time/resource constraints.
 4. False. See the answer to the option below, which was the modified version of the question actually used in the quiz. This question is included here as a slightly different version that may be a more intuitive way to think of the question, and to understand why the encoder loss in the VAE is not easy to optimize, as mentioned in the last slide of the relevant lecture.
 5. True. The reconstruction loss is dependent on the encoder parameters, as we first sample from the latent space and then try to see if we can reproduce the original distribution from those latent samples. If we had optimized the encoder loss greedily by setting $\mu_{z|x^i} = 1, \sigma_{z|x^i} = 1$, the reconstruction loss would blow up as it would be hard to reproduce x from truly random noise. A DDPM avoids this by making noise the final step, having many intervening steps where a small set of features can be generated.
3. Consider a 3 variable markov random field with 2 cliques ($\{x_1, x_2\}, \{x_1, x_3\}$), whose parameters we want to estimate via the EM algorithm. Generally, there would be 8 parameters to consider and maximize over in the M step, where we compute

$$\arg \max_{\theta} \sum_i \sum_{x_1} q_t(x_1 | x_2, x_3) P(x_1, x_2, x_3 | \theta)$$

But, if we know that every potential table is of the form $\phi_{ij}(0, 0) = \phi_{ij}(1, 1) = \alpha$, $\phi_{ij}(0, 1) = \phi_{ij}(1, 0) = 1 - \alpha$, then there is only one parameter to estimate. Suppose we observe two datapoints $\{(x_2 = 1, x_3 = 1), (x_2 = 0, x_3 = 1)\}$. Assume our initial estimate for α is $\frac{3}{4}$

1. Run the E step of the EM algorithm. Let $q(x_1 = 1|x_2 = 1, x_3 = 1) = \frac{a}{b}$ for coprime a, b. Find a+b.
2. Run the E step of the EM algorithm. Let $q(x_1 = 1|x_2 = 0, x_3 = 1) = \frac{a}{b}$ for coprime a, b. Find a+b.
3. Run the M step of the EM algorithm after the E step to update your belief of α . Let the new value of α you get be $\frac{a}{b}$, for coprime a,b. Find a+b.
4. Realizing that even without the EM algorithm, finding an optimal value of α might be possible through arithmetic, Alice wants to compute a value of α (not equal to 0 or 1) such that, if the EM algorithm is initialized with that α , after the M step she would get α again. Find the value of α that satisfies this property. Let it be $\frac{a}{b}$ for coprime a, b. Find a+b.

Solution:

1. $\frac{9}{10}$. a+b=19. Consider $q(x_1 = 1|x_2 = 1, x_3 = 1) \propto \phi_{12}(1,1)\phi_{13}(1,1) = \frac{9}{16}$, $q(x_1 = 0|x_2 = 1, x_3 = 1) \propto \phi_{12}(0,1)\phi_{13}(0,1) = \frac{1}{16}$. $q(x_1 = 1|x_2 = 1, x_3 = 1) = \frac{9}{9+1}$
 2. $\frac{1}{2}$. a+b=3. Consider $q(x_1 = 1|x_2 = 0, x_3 = 1) \propto \phi_{12}(1,0)\phi_{13}(1,1) = \frac{3}{4}$, $q(x_1 = 0|x_2 = 0, x_3 = 1) \propto \phi_{12}(0,0)\phi_{13}(0,1) = \frac{3}{4}$. $q(x_1 = 1|x_2 = 1, x_3 = 1) = \frac{3}{3+3}$
 3. $\frac{9}{10} \log \alpha^2 + \frac{1}{10} \log (1 - \alpha)^2 + (\frac{1}{2} + \frac{1}{2}) \log \alpha(1 - \alpha) - 2 \log Z$, where Z is 2 irrespective of α . $\alpha = \frac{28}{40} = \frac{7}{10}$.
 4. A+B+C=8. One gets A=3, B=1, C=4. $\frac{2\alpha^2}{\alpha^2+(1-\alpha)^2} \log \alpha + \frac{2(1-\alpha)^2}{\alpha^2+(1-\alpha)^2} \log (1 - \alpha) + \log \alpha(1 - \alpha)$ is the function to be minimized. $3x^2 + (1 - x)^2 - 4x(x^2 + (1 - x)^2) = 0$. This is satisfied by $x = \frac{1}{2}$
4. Consider a noise schedule $\bar{\alpha}_t = 1 - \left(\frac{t}{T}\right)^2$, where $T = 50$.
1. Find the value of β_t at $t = 26$, let this value be $\frac{a}{b}$, where a and b are coprime natural numbers. Find a+b.
 2. Suppose we are trying to predict a two-dimensional variable $x = [a, b]$; thus the diffusion process takes $x_0 = [a_0, b_0]$ and generates $x_1 = [a_1, b_1], x_2 = [a_2, b_2]$ and so on. What is the value of $\text{Var}(a_{25} - b_{25})$ if $[a_0, b_0]^T \sim \mathcal{N}(\mu = [1, 1]^T, \Sigma = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix})$, where $\text{Var}(x)$ denotes the variance of the RV x? Let this answer be $\frac{a}{b}$, where a, b are coprime natural numbers. Find a+b.

Solution:

1. $\beta_t = 1 - \frac{\bar{\alpha}_t}{\bar{\alpha}_{t-1}}$. Here, this is $\frac{26^2-25^2}{50^2-25^2} = \frac{17}{625}$. Thus, the answer is 17+625=642.
2. $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$, where ϵ is an $N(0, I_2)$ Gaussian. The Covariance matrix is thus $\bar{\alpha}_t \Sigma + (1 - \bar{\alpha}_t)I_2$. $\text{Var}(a_{25} - b_{25}) = \text{Var}(a_{25}) + \text{Var}(b_{25}) - 2\text{Covar}(a_{25}, b_{25}) = \frac{3}{4} * 3 + \frac{1}{4} + \frac{3}{4} * 2 + \frac{1}{4} - 0 = \frac{17}{4}$. The answer is thus 21.